Example 1.1

Suppose that a population consists of 4 elements such as 4, 8, 12 and 16. These may be considered as the values of a random variable, say, X. Let a random sample of size 2 be drawn from this population under *sampling with replacement* scheme. Then, the possible number of samples is 4^2.

It is to be noted that, if we take samples of size n each from a finite population of size N, then the number of samples will be N^n under with replacement scheme and ${}^{N}C_n$ samples under without replacement scheme.

In each of the 4^2 samples, the sample elements x_1 and x_2 can be considered as the values of the two *iid* random variables X_1 and X_2. The possible samples, which could be drawn from the above population and their respective means are presented in Table 1.2.

Table 1.2 Possible Samples and their Means

Sample Number	Sample elements (x_1, x_2)	Sample Mean $\bar{x}$
1	4,4	4
2	4,8	6
3	4,12	8
4	4,16	10
5	8,4	6
6	8,8	8
7	8,12	10
8	8,16	12
9	12,4	8
10	12,8	10
11	12,12	12
12	12,16	14
13	16,4	10
14	16,8	12
15	16,12	14
16	16,16	16

The set of pairs (x_1, x_2) listed in column 2 constitute the sample space of samples of size 2 each.

Hence, the sample space is:

$\mathbf{S}$ = {(4,4), (4,8), (4,12), (4,16), (8,4), (8,8), (8,12), (8,16), (12,4), (12,8), (12,12), (12,16), (16,4), (16,8), (16,12), (16,16)}

The sampling distribution of $\overline{X}$, the sample mean, is determined and is presented in Table 1.3.

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